

their competence for a given analyte or method, in terms of trueness. They are routinely used by accredited laboratories, which are summoned to participate at several proficiency testing scheme (PTS) per year to remain accredited. More details are available in this chapter.

It is quite different of the classic interlaboratory ring study described in Section 3.1 about precision. It brings together only some laboratories which must perform replicates. It is mainly devoted to estimating the repeatability and reproducibility of a method, but ISO 5725 standards also describe several extensions and parameters relating to trueness. The major disadvantage is the cost in such a way that interlaboratory studies are not routinely used for this specific purpose. This topic is not addressed here.

4.3.2 Organization of Proficiency Testing Schemes

While ISO 5725 interlaboratory analyses are organized at the initiative of a limited group of laboratories that want to standardize an operating procedure according to standard recommendations, PTSs are managed by specialized organizations that follow the rules available in ISO standards and guides:

Reference	Topics covered
ISO/IEC Guide 43-1	Implementation and organization of PTS
ISO/IEC Guide 43-2	Relationship between PTS and certification bodies
ISO 13528	Statistical methods for the interpretation of PTS
ISO 17043	Requirements for PTS organizers

Usually, the legal status of the managing organization is nonprofit and issued by a professional association, such as cereal products or clinical biology. But it can also be a governmental body if regulatory purposes are involved. In some cases, participation in a PTS is a regulatory requirement. For example, biomedical analysis laboratories must participate in quality control (QC) programs to keep their right to make analyses on humans. To the extent that organizers can obtain certification, it is essential that they have no conflict of interest with participants.

There are many PTS organizers around the world. At the European level, the European Proficiency Testing Schemes (EPTIS) is managing a database available on the site <http://www.eptis.bam.de> that lists several hundred PTSs in all fields of analysis. Similarly, the US National Association for Proficiency Testing (<https://proficiency.org/>) and Proficiency Testing Canada (<https://ptcanada.org/>) play the role of PTS providers in their own countries. The participants can be several hundred or thousands, usually contributors to the nonprofit managing organization. In that context they participate in PTS items selection and interpretation procedures. Each year they receive one or several test samples that they measure with their own routine method. One single measurement value is required but it is not a general rule.

One PTS often covers several analytes. The results are sent back to the organizer, who will calculate a reference value with acceptance limits. The participating laboratory must be within these limits to be declared competent, as required by several accreditation bodies. Finally, a PTS can be summarized in four steps:

1. Select and prepare a homogenized test material packaged in adapted containers. It is usual that they only have the quantity sufficient for the expected number of measurements.
2. Send one item to each member of a panel of subscribing laboratories.
3. Estimate the assigned reference value with its acceptance limits according to one of the consensus calculation methods described in the next section.
4. Compute a score for each participating laboratory and decide whether it is acceptable; the agreement depends on the value of the score.

The principle is therefore quite simple but the organizational and logistical aspects are crucial, justifying the implementation of a quality system for PTS organizers. Especially since they are service providers for accredited laboratories. The organizer's competence mainly lies in the ability to produce homogeneous test materials. Although this is an obvious technical problem, it is impossible to give general rules here, given the variety of matrix types available and the types of analytes. Some materials combined with certain analytes can be very unstable, such as microorganisms in water or foods, while others, such as lead isotopes in rocks, are very stable.

Therefore, any conceivable preservation technology can be used: drying, freeze-drying, canning, pasteurization, or other as well as all types of packaging: inactive, vacuum, controlled atmosphere, and so on. Most materials are shipped in such quantities so as to avoid the laboratory making additional replicates and no longer remaining in routine conditions. Often, once the package is opened, the contents can change, and the material should not be reused. In addition to the planning aspects, the interpretation of proficiency tests raises statistical problems at three levels:

1. Select an algorithm to estimate the reference value that will be assigned to the material at the end of the PTS. This question of defining the central value of sampled data is as old as statistical science itself. Since this material can eventually be used as an external RM, it is also important to know how to express the MU.
2. Check material homogeneity and stability. If the material is not homogeneous, there is a risk that a laboratory's result will be interpreted as being due to its incompetence. The recommended method is to achieve homogeneity checking at the beginning, but it can be complemented by using the results the laboratories sent back.
3. Define a score of competence for a laboratory. The aim is to establish a certificate that can be provided to an inspector in charge of auditing the laboratory.

4.3.3 Reference Value of the Test Material

Some years ago, the influence of various estimators used by different organizers on laboratory scoring was demonstrated [1]. The situation has changed since

Table 4.1 Terms used in ISO 13528.

Assigned value	X	Value attributed to a particular property of a proficiency test item. It is also referred to as the reference value.
Standard deviation for proficiency assessment	s_A	The measure of dispersion used in the evaluation of the results of proficiency testing based on the available information. It is comparable to a standard deviation interpreted as a target dispersion value for a population of laboratories.

already-quoted international standards were established. Statistical data processing is no longer left to the free choice of the test organizers, although the proposed estimators are flexible. Two new concepts were introduced in the standards, as defined in Table 4.1.

Whereas ISO 13528 is applied in many fields of measuring sciences, the definitions are confusing as they try to merge different methodological frameworks. Five estimators can be used for assigning the reference value X to the test material. Each method comes with an estimate of the standard-uncertainty $u(X)$.

The Table 4.2 summarizes these methods. It is assumed that I laboratories are participating in the PTS with $1 < i < I$.

The robust estimates obtained by algorithm A are described in Section 4.3.4. The reason for robust estimators in PTS to be proposed can be explained as no laboratory can be rejected as an outlier, since it is compulsory to give a score to every participant,

Table 4.2 Estimators of the assigned value and their standard-uncertainty.

Method	Nature of the test material	Standard-uncertainty
Formulation	Surrogate matrix or standard addition	Law of propagation of uncertainty (Section 6.6)
Certified value	Certified reference material Require simultaneous measures of X_i and C_i on the test material and CRM, respectively. Calculate the average difference.	$u(X) = u(X_{CRM})$
Calibration by CRM	$\bar{D} = \frac{\sum_i (X_i - C_i)}{I}$ $X = X_{CRM} + \bar{D}$	$u(X) = \sqrt{u^2(X_{CRM}) + u^2(\bar{D})}$
Expert laboratories	A group of I' expert laboratories provide X and their uncertainty $u(X_i)$ with $1 < i < I'$	$u(X) = \frac{1.23}{I' \sqrt{\sum_i u^2(X_i)}}$
All laboratories	Algorithm A: X robust average and s_A^* robust standard deviation (Section 4.3.5)	$u(X) = \frac{1.23 \times s_A^*}{\sqrt{I}}$

Table 4.3 Estimators of the standard deviation for proficiency assessment.

Method	Principle	Formula
Prescribed value	Regulatory requirement	Not applicable
From a model	e.g. the Horwitz's model	$s_A = 0.02X^{-0.849}$
By perception	Prerequisite performance level	$s_A = \frac{X}{3}$
Interlaboratory study	Based on s_r, s_R	$s_A = \sqrt{2.25 (s_R^2 - s_r^2) + \frac{s_r^2}{I}}$
Proficiency testing	Algorithm A (Section 4.3.5)	s_A^*

even when it is an obvious outlier. Moreover, statistical testing for outlier rejection always assumes a simple distribution function behind the dataset, usually Normal or unimodal. Experience shows that the situation is more complex as bimodal or more distributions may occur.

For the standard deviation s_A , five methods are also available and reviewed in Table 4.3, including the robust calculation with algorithm A described below. The organizer must choose among these methods, in consultation with the members of the PTS, an eventual accreditation body, or the regulatory requirements.

4.3.4 Performance Scores

A performance score is applied to measure the extent to which a given laboratory diverges from the others and to judge whether it has the same competence as the rest of the group of laboratories that participated in the test. Therefore, a score must be accompanied by an acceptance interval to allow this judgment. Table 4.4 combines the five calculation methods proposed by the ISO standards, with their acceptance limits. The first two scores are just usual measures of bias. The z -score is the best-known and most widely used. It consists of centering the result obtained (by

Table 4.4 The different scores available for PTS.

Designation	Formula	Acceptance
Absolute bias	$D_i = Z_i - X$	$\pm 3 s_A$
Relative bias	$D\%_i = \frac{Z_i - X}{X} \times 100$	$\pm 3 \frac{X}{s_A}$
z -score	$z_i = \frac{Z_i - X}{s_A}$	± 3.0
Standardized deviation EN	$EN_i = \frac{Z_i - X}{\sqrt{U^2(Z_i) + U^2(X)}}$	± 1.0
z' -score	$z'_i = \frac{Z_i - X}{\sqrt{s_A^2 + u^2(X)}}$	± 3.0

subtracting the mean) and reducing it (by dividing it by the standard deviation) to find a centered and reduced Normal distribution. The advantage of this operation called *standardization* is to avoid scale effects that exist with absolute bias and allows comparison of the laboratory's performance for various tests, regardless of the units used or the concentration level.

Understanding the principles used to determine the acceptance interval does not require many comments. For example, for the z -score, it is referring to the standard Normal distribution law, and ± 3 limits correspond to a probability interval of 99.7%. Proposed score must be interpreted separately, in the framework of a given PTS. When a laboratory participates in several PTS, it would be an error to compute some average value because each score is linked to a specific set of values.

4.3.5 Algorithm A

The existence of robust statistics, as defined by statisticians, was already introduced in Section 3.4.2 about the possible presence of outliers. The theory of this new part of statistics was developed in the 1980s, and many robust parameters have been published since. They are good reference books, but this is not the right place to review possible proposals [2]. The aim is to only give an insight into a robust estimation method, introduced by ISO, in relation to the assessment of the PTS performance. The results of a proficiency test that involved determining moisture in alfalfa are used to illustrate the interest in so-called algorithm A, which simultaneously gives robust estimates of mean and standard deviation. Thirty-six laboratories, labeled L01 to L36, participated and their results are listed in Table 4.5.

Algorithm A is a full part of the ISO 13528 standard. Resource G explains how the algorithm works and how robust parameters are iteratively computed until a convergence criterion is reached. Because it is an iterative algorithm, it is simpler to use a Python script but it is programmable in Excel with Visual Basic for Applications.

Table 4.5 ALFALFA^{a)} – proficiency testing of moisture determination (g/100 g).

Lab	Z_i	Lab	Z_i	Lab	Z_i	Lab	Z_i
L01	7.59	L10	8.06	L19	8.17	L28	8.27
L02	7.79	L11	8.07	L20	8.18	L29	8.28
L03	7.81	L12	8.07	L21	8.19	L30	8.29
L04	7.81	L13	8.10	L22	8.20	L31	8.29
L05	7.84	L14	8.11	L23	8.21	L32	8.30
L06	7.86	L15	8.15	L24	8.22	L33	8.31
L07	7.92	L16	8.15	L25	8.23	L34	8.31
L08	7.96	L17	8.15	L26	8.27	L35	8.36
L09	7.99	L18	8.16	L27	8.27	L36	8.57

a) Unpublished personal data.